memory error: VQE simulation of H2O

https://github.com/qiskit-community/qiskit-aqua/issues/1108

ROW 38

memory error: VQE simulation of H2O #1108

Closed

ares201005 opened on Jul 9, 2020

Information

- **Qiskit Aqua version**: 0.7.3
- **Python version**: 3.7
- **Operating system**: Linux

What is the current behavior?

The VQE simulation of H2o molecule results in memory error:

.../site-packages/qiskit/aqua/operators/primitive_ops/pauli_op.py
../site-packages/qiskit/quantum_info/operators/pauli.py
.../site-packages/scipy/sparse/compressed.py
.../site-packages/scipy/sparse/base.py
MemoryError: Unable to allocate 4.00 GiB for an array with shape (16384, 16384) and data type complex128

However, my computer has enough memory, my other script reqesting much larger memory still works fine.

Steps to reproduce the problem

using H2O molecule, VQE with UCCSD ansatz, L_BFGS_B optimizer,

What is the expected behavior?

With earlier version of qiskit-aqua (0.6.5) VQE simulation of the H2O molecule does not have such a problem.

Suggested solutions

woodsp-ibm on Jul 9, 2020 Member

Are you using statevector_simulator? If so this is most likely the same issue as #1027 and in which case might I suggest you use the Aer qasm_simulator and set `use_custom` to True on VQE (see https://qiskit.org/documentation/stubs/qiskit.aqua.algorithms.VQE.html for docs on this). This will use the custom Aer snapshot instruction to compute the expectation (uses AerPauliExpectation object inside VQE to do this - though you can pass this in explicitly to VQE). Overall this mode is more efficient too than running with statevector and like statevector has the ideal outcome without shot noise.

ares201005 on Jul 10, 2020 Author

Yes, it is a statevector_simulator. Thank you for the hint, it indeed solves the problem.

adekusar-drl closed this as completed on Jul 14, 2020

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